than are available at present. But we found that with those capital market expectations, the 4 percent retirement withdrawal rate strategy may only be appropriate for more risk-averse retirees with moderate guaranteed income sources. The ability to accept greater failure probabilities means that risk-tolerant retirees will prefer a higher withdrawal rate and a riskier retirement portfolio. A risk-tolerant retiree may prefer a withdrawal rate of between 5 and 7 percent with a guaranteed income of $20,000. As well, the optimal allocation to stocks increases by between ten and thirty percentage points, and the optimal withdrawal rate increases by between one and two percentage points for retirees with a guaranteed income of $60,000 instead of $20,000.

Again, there is an important point to reemphasize here. In one case in the article, we identify a 7 percent withdrawal rate as “optimal.” That is not a “safe” withdrawal rate. With the market assumptions in the article, the 7 percent withdrawal rate has a 57 percent chance of failure over a thirty-year retirement. Though it is not safe, it does maximize the overall expected lifetime satisfaction for a fairly flexible retired couple who has a secured income base of $20,000 from Social Security. In our analysis, 7 percent is how the couple could best balance the trade-off between spending more in early retirement and the possibility of then having to spend less later in retirement.

In April 2017, David Blanchett published an article in the *Journal of Financial Planning* titled “The Impact of Guaranteed Income and Dynamic Withdrawals on Safe Initial Withdrawal Rates.” His article provides more recent further confirmation for this discussion, and his objective was to quantify the relative impact of different factors on initial spending rates and acceptable probabilities of failure. He investigates how optimal initial withdrawal rates are impacted by the amount of guaranteed income, the extent that the household can adjust spending, the risk of the investment portfolio, the return assumptions used for projections, and the degree of income stability desired by the retiree.

Blanchett finds that the level of guaranteed income is by far the most important factor in explaining optimal spending rates from investments. Withdrawal rates can be up to four percentage points higher across the